Strengths of Acids and Bases

Strong vs. weak, and the equilibrium constants that measure it

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Standards

Chemistry 2.6 — Acid–Base Chemistry: strength, ionization equilibrium, Ka and Kb.

Objectives

1. What distinguishes a strong acid/base from a weak one?
2. How does Ka (or Kb) quantify acid (or base) strength?
3. Why does a strong acid have a weak conjugate base?

Expectations

1. Classify acids/bases as strong or weak from their equations
2. Compare strength using Ka and Kb values

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What Do You Already Know?

WHAT I KNOW	WHAT I WANT TO KNOW	EXPLORE THE SOURCE
▪ Car batteries contain a strong acid ▪ Baking soda is a mild, weak base ▪ Vinegar is a weak acid, safe on food	▪ Why does a weak acid still "count" as an acid? ▪ How small can Ka actually get? ▪ Does strong always mean dangerous?	► PhET: Acid-Base Solutions ► PhET: pH Scale ► Khan Academy: Acids & Bases *Tap a link to explore before class discussion.*

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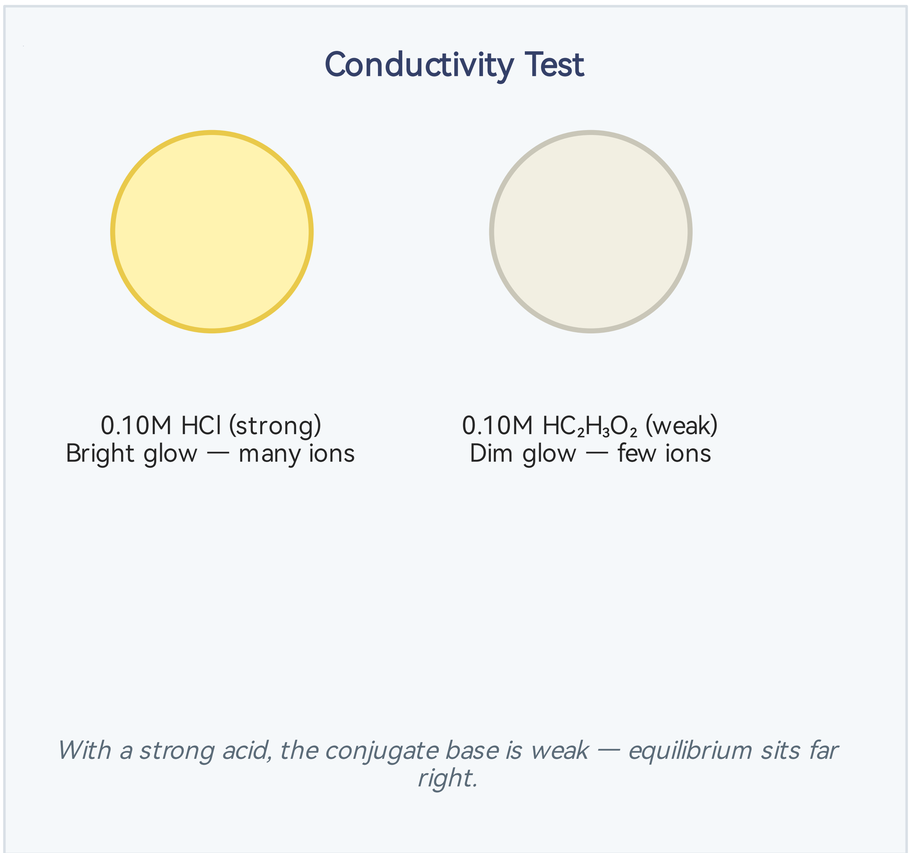
Strengths of Acids

Strong acids

- Ionize completely: $HCl(aq) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$
- Maximum H^+ produced → excellent conductors of electricity

Weak acids

- Ionize only partially: $HC_2H_3O_2 \rightleftharpoons H^+ + C_2H_3O_2^-$
- Fewer free ions → dim glow / poor conductor
- Analogy: a strong acid is a crowd that lets go of the ball all at once; a weak acid is a crowd where only a few ever let go.



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Strong Acids vs. Weak Acids

STRONG ACIDS	
Acid	Ionization
Hydrochloric	$HCl \rightarrow H^+ + Cl^-$
Hydroiodic	$HI \rightarrow H^+ + I^-$
Perchloric	$HClO_4 \rightarrow H^+ + ClO_4^-$
Nitric	$HNO_3 \rightarrow H^+ + NO_3^-$
Sulfuric	$H_2SO_4 \rightarrow H^+ + HSO_4^-$

WEAK ACIDS	
Acid	Ionization
Hydrofluoric	$HF \rightleftharpoons H^+ + F^-$
Acetic	$HC_2H_3O_2 \rightleftharpoons H^+ + C_2H_3O_2^-$
Hydrosulfuric	$H_2S \rightleftharpoons H^+ + HS^-$
Carbonic	$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$
Hypochlorous	$HClO \rightleftharpoons H^+ + ClO^-$

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The Acid Ionization Constant, Ka

Acid	Ionization Equation	Ka (298 K)
Hydrofluoric	$HF \rightleftharpoons H^+ + F^-$	6.3×10^{-4}
Acetic	$CH_3COOH \rightleftharpoons H^+ + CH_3COO^-$	1.8×10^{-5}
Carbonic (1st)	$H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$	4.5×10^{-7}
Hydrosulfuric (1st)	$H_2S \rightleftharpoons H^+ + HS^-$	8.9×10^{-8}
Hydrocyanic	$HCN \rightleftharpoons H^+ + CN^-$	6.2×10^{-10}

Ka = $[H^+][A^-] / [HA]$. A smaller Ka means a weaker acid.

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Strengths of Bases

Base	Ionization Equation	Kb (298 K)
Ethylamine	$C_2H_5NH_2 + H_2O \rightleftharpoons C_2H_5NH_3^+ + OH^-$	5.0×10^{-4}
Methylamine	$CH_3NH_2 + H_2O \rightleftharpoons CH_3NH_3^+ + OH^-$	4.3×10^{-4}
Ammonia	$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	2.5×10^{-5}
Aniline	$C_6H_5NH_2 + H_2O \rightleftharpoons C_6H_5NH_3^+ + OH^-$	4.3×10^{-10}

Strong bases (NaOH, KOH, $Ca(OH)_2$, etc.) dissociate completely and don't appear in a Kb table.

conjugate acid of the Base. NH_3

$H^+ + NH_3 \rightleftharpoons NH_4^+$

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Lesson Summary

Key Takeaways

1. Strong acids/bases ionize completely; weak acids/bases ionize only partially and exist mostly as neutral molecules at equilibrium.

2. Strong electrolytes conduct electricity well because they release the most ions.

3. Ka and Kb quantify strength: the smaller the constant, the weaker the acid or base.

4. With a strong acid, the conjugate base is weak (and vice versa) — strength trades off across the conjugate pair.

Test Yourself

1. Write ionization equations and Ka expressions for $HClO_2$, HNO_2, and HIO.

2. Write the first and second ionization equations for H_2SeO_3.

3. Calculate the Ka of a 0.00330M solution of benzoic acid given pOH = 10.70.

4. Calculate the Ka of a 0.100M solution of cyanic acid (HCNO) given pOH = 11.00.

5. Challenge: given Ka = $[AsO_4^{3-}][H_3O^+] / [HAsO_4^{2-}]$, write the balanced equation for the reaction.

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